

MAT3280 Final exam – sample solution

Question 1. (10 points) Let r be a positive integer and p a real number between 0 and 1. The negative binomial distribution has probability mass function defined by

$$p(k) = \binom{k+r-1}{k} p^r (1-p)^k,$$

for $k = 0, 1, 2, \dots$. We denote this distribution by $NegBin(r, p)$.

(a) Compute the moment generating function of $NegBin(r, p)$.

(Hint: Use the power series expansion

$$(1+x)^{-n} = \sum_{k=0}^{\infty} \frac{(-n)(-n-1)\cdots(-n-k+1)}{k!} x^k$$

for positive integer n .)

(b) Let X and Y be independent variable, with X distributed according to $NegBin(r, p)$ and Y distributed according to $NegBin(s, p)$, where r and s are positive integers. Show that the distribution of $X + Y$ is negative binomial.

Solution. Suppose $X \sim NegBin(r, p)$. We want to calculate the moment generating function of X

$$m_X(t) = \sum_{k=0}^{\infty} \binom{k+r-1}{k} p^r (1-p)^k e^{tk}$$

From the power series in the hint, we have

$$\begin{aligned} (1+x)^{-r} &= \sum_{k=0}^{\infty} \frac{(-r)(-r-1)\cdots(-r-k+1)}{k!} x^k \\ &= \sum_{k=0}^{\infty} \frac{r(r+1)\cdots(r+k-1)}{k!} (-x)^k \\ &= \sum_{k=0}^{\infty} \binom{k+r-1}{k} (-x)^k, \end{aligned}$$

for x within the radius of convergence. Using this power series expansion, we can write

$$m_X(t) = p^r \sum_{k=0}^{\infty} \binom{k+r-1}{k} [(1-p)e^t]^k = \frac{p^r}{(1-(1-p)e^t)^r}.$$

(b) The moment generating functions of X and Y are

$$\frac{p^r}{(1 - (1 - p)e^t)^r} \text{ and } \frac{p^s}{(1 - (1 - p)e^t)^s},$$

respectively. Since X and Y are independent, the moment generating function of $X + Y$ is the product

$$\frac{p^r}{(1 - (1 - p)e^t)^r} \cdot \frac{p^s}{(1 - (1 - p)e^t)^s} = \frac{p^{r+s}}{(1 - (1 - p)e^t)^{r+s}}.$$

This is the moment generating function of $NegBin(r + s, p)$. Since moment generating function uniquely determine the underlying distribution, we can conclude that $X + Y$ is distributed according to the negative binomial distribution with parameters $r + s$ and p .

(The analog in the world of continuous distribution is the Gamma distribution. It is known that the sum of two independent Gamma distributions is another Gamma distribution.)

Question 2. (15 points) Let X denote a random variable uniformly distributed between -1 and 1 , and let $Y = X^2$.

(a) Find the best linear estimator of X given Y . That is, find a constant k such that the mean square error between X and kY is minimized.

(b) Find the best nonlinear estimator of X given Y . That is, find the conditional expectation $E[X|Y]$. (You can briefly explain your answer by highlighting the main points. A complete proof using the π - λ theorem is not required.)

Solution. We want to minimize a quadratic function in variable k :

$$E[(X - kY)^2] = E[X^2] - 2kE[XY] + k^2E[Y^2].$$

The minimum value is attained if we set

$$k = \frac{E[XY]}{E[Y^2]}.$$

We next calculate $E[XY]$ and $E[Y^2]$.

$$\begin{aligned} E[XY] &= \int_{-1}^1 \frac{1}{2} x(x^2) dx = 0. \\ E[Y^2] &= \int_{-1}^1 \frac{1}{2} (x^2)^2 dx \\ &= \int_0^1 x^4 dx \\ &= \frac{1}{5} [x^5]_0^1 = \frac{1}{5}. \end{aligned}$$

Therefore, the optimal value of k is $0/(1/5)=0$. The best estimate of X given Y is 0 .

(b) In this example, even if we allow nonlinear function, the best estimator of X given Y is also the constant 0 . Given that we know Y is equal to a number y , X can take only two values, $\sqrt{y}$ and $-\sqrt{y}$, and both of them are equally probable. As a result, the expectation is 0 for any given y .

The argument in the previous paragraph technically is not very correct, because it involves the conditioning on an event with zero probability, incurring the error of division by zero. To be more rigorous, we consider an interval $[a, b]$, for $0 < a < b < 1$, whose width is nonzero. The inverse image of $[a, b]$ under the square function $f(x) = x^2$ is the union

$$B \triangleq [-\sqrt{b}, -\sqrt{a}] \cup [\sqrt{a}, \sqrt{b}]. \quad (1)$$

The set B is $\sigma(Y)$ -measurable.

The integral of X under B is

$$\int_B X dP = \int_{-\sqrt{b}}^{-\sqrt{a}} x dx + \int_{\sqrt{a}}^{\sqrt{b}} x dx = 0.$$

Therefore for any set B in the form of (1), we have

$$\int_B x dx = \int_B 0 dx.$$

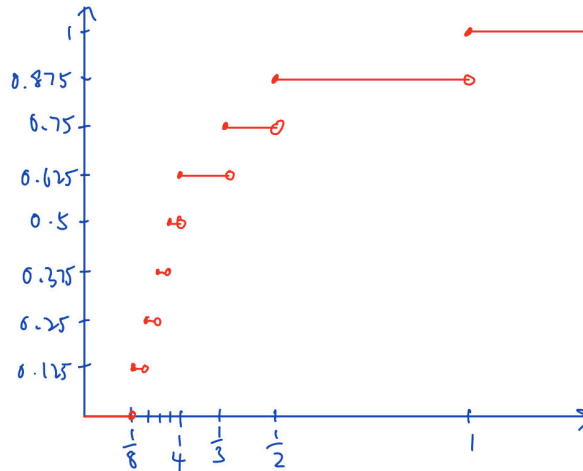
This proves that $E[X|Y] = 0$ a.s. because sets in the form in (1) generate $\sigma(Y)$.

Question 3. (15 points) For positive integer n , let X_n be the discrete random variable uniformly distributed on the integers from 1 to n , i.e., $P(X_n = i) = 1/n$ for $i = 1, 2, \dots, n$. Let $Y_n = 1/X_n$.

(a) Sketch the cumulative distribution function of Y_n .

(b) Does the sequence of random variables $(Y_n)_{n=1}^{\infty}$ converge in distribution? If the answer is yes, what is the limiting distribution?

Solution. (a) An example of cdf for Y_8 is shown below



For general n , we can describe the cdf of Y_n as

$$F_n(x) \triangleq P(Y_n \leq x) = \begin{cases} 1 & \text{if } x \geq 1, \\ (n-1)/n & \text{if } 1/2 \leq x < 1, \\ (n-2)/n & \text{if } 1/3 \leq x < 1/2, \\ (n-3)/n & \text{if } 1/4 \leq x < 1/3, \\ \vdots & \vdots \\ 1/n & \text{if } 1/n \leq x < 1/(n-1), \\ 0 & \text{if } x < 1/n. \end{cases}$$

(b) The cdf of the random variables Y_n 's converge to the step function $\mathbf{1}_{\{x \geq 0\}}$, for $x > 0$. This is the cdf of the constant random variable 0.

In fact, we can see that $F_n(x)$ is constant for $1/2 \leq x < 1$ is equal to $1 - 1/n$, which converges to 1 as $n \rightarrow \infty$. For x in the range $[1/3, 1/2)$, $F_n(x)$ is equal to $1 - 2/n$, which converges to 1 as $n \rightarrow \infty$. In general, we divide the interval $(0, 1)$ into the disjoint union of intervals in the form $[1/k, 1/(k-1))$, and verify that $F_n(x)$ is constant in each of these intervals, and the value of $F_n(x)$ converges to 1 for $x \in [1/k, 1/(k-1))$. Hence, as $n \rightarrow \infty$, the sequence of functions $F_n(x)$ converges to

$$F(x) \triangleq \begin{cases} 1 & \text{if } x \geq 0, \\ 0 & \text{if } x < 0 \end{cases}$$

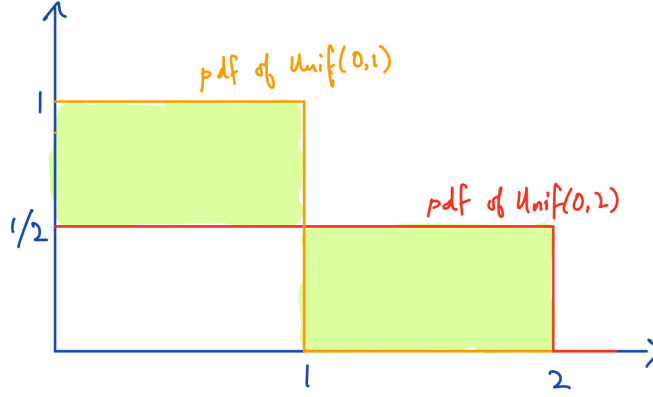
for all x except the discontinuity point $x = 0$ of $F(x)$. Hence, Y_n converges to the constant random variable 0 as $n \rightarrow \infty$.

Question 4. (15 points) We use the name $Unif(a, b)$ to denote the continuous uniform distribution between a and b in this question.

(a) Find the total variation distance between $Unif(0, 1)$ and $Unif(0, 2)$.

(b) Describe a way to generate two random variables X and Y , such that $X \sim Unif(0, 1)$, $Y \sim Unif(0, 2)$, such that the probability $P(X \neq Y)$ is equal to the total variation distance between $Unif(0, 1)$ and $Unif(0, 2)$. (Hint: Use a coin toss and a uniform random variable as the random source. Write down a deterministic function that transforms them to a pair of random variables (X, Y) .)

Solution. (a) The total variation distance is equal to the area between the two probability distribution functions, divided by 2.



The answer is equal to $1/2$.

If we want to put it more formally, we can first write the pdf of $Unif(0,1)$ and $Unif(0,2)$ as

$$f(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise,} \end{cases}$$

$$g(x) = \begin{cases} 1/2 & \text{if } 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

The TV distance between $f(x)$ and $g(x)$ is

$$d_{TV}(f, g) = \frac{1}{2} \int_0^2 |f(x) - g(x)| dx = \frac{1}{2} \int_0^2 (1/2) dx = 1/2.$$

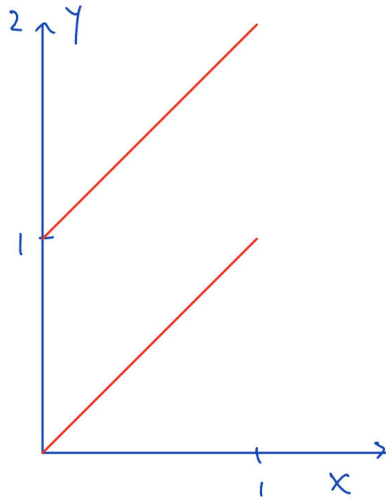
(b) We toss a fair coin and a toss and generate a uniform random variable U independently. If the outcome of the coin is head, we let $X = U$ and $Y = U$. Otherwise if the outcome of the coin is tail, we let $X = U$ and $Y = 1 + U$.

No matter what the outcome of coin is, X is equal to U , and hence X is uniformly distributed between 0 and 1. For Y , with probability $1/2$ it is distributed between 0 and 1, and with probability $1/2$, it is distributed between 1 and 2. Overall, Y is distributed between 0 and 2 uniformly. When the coin toss is head, then $X = Y$, otherwise, $X \neq Y$. The probability that $X \neq Y$ is equal to $1/2$, meeting the total variation distance between $Unif(0,1)$ and $Unif(0,2)$.

The joint distribution of X and Y is an example of singular distribution. A random sample will fall on the lines $x = y$ or $x = y + 1$, with x in the range $0 \leq x \leq 1$. The samples will only occur in a set with zero area. Strictly speaking the pdf of this distribution does not exist. However, if we use the Dirac delta function (which is a kind of generalized function), we can express the pdf as

$$\frac{1}{2} \delta(x - y) + \frac{1}{2} \delta(x - y - 1)$$

for $0 \leq x \leq 1$. The joint pdf is illustrated below. All probabilities are concentrated on the two line segments drawn in red color.



We can understand the answer by implementing it in Python. We generate two independent uniform random variables as the random source. The result of coin toss is determined from the first uniform random variables — head if less than $1/2$ and tail if larger than $1/2$.

```
from numpy import random
U1 = random.uniform()
U2 = random.uniform()

if U1 < 0.5:
    X=Y=U2
else:
    X=U2; Y=U2+1

print(f"(X,Y) = ({X},{Y})")
```

Sample outputs:

```
(X,Y) = (0.3448500871960185,1.3448500871960185)
(X,Y) = (0.8897828781739293,0.8897828781739293)
(X,Y) = (0.7682760229207967,0.7682760229207967)
(X,Y) = (0.871160569228811,1.871160569228811)
```

If we take a lot of samples by repeatedly running this program, approximately half of the samples have $X = Y$ and the other half satisfies $X \neq Y$. The probability that $X \neq Y$ is $1/2$, achieving the total variation distance between X and Y .

Question 5. (15 points) Let $(X_n)_{n=1}^{\infty}$ be a sequence of nonnegative i.i.d. random variable with finite mean. Let the common mean be denoted by μ .

(a) For any positive real number ϵ and positive integer n , define the indicator function $\mathbf{1}_{\{x > \epsilon n\}}$, which is equal to 1 if $x > \epsilon n$, and equal to 0 otherwise. Show that, for fixed $\epsilon > 0$, the following inequality holds for real number $x \geq 0$,

$$\sum_{n=1}^{\infty} \mathbf{1}_{\{x > \epsilon n\}} \leq \frac{x}{\epsilon}.$$

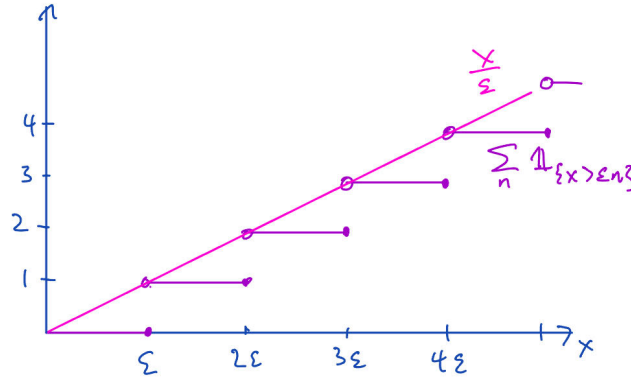
(Hint: sketch the summation on the left hand side as a function of x , and check that this is dominated by the linear function on the right.)

(b) Using the inequality in part (a), show that for any $\epsilon > 0$,

$$\sum_{n=1}^{\infty} P\left(\frac{X_n}{n} > \epsilon\right) < \frac{\mu}{\epsilon},$$

and hence show that the sequence $(X_n/n)_{n=1}^{\infty}$ converges to 0 almost surely.

Solution. (a) Let ϵ be a fixed positive real number. The summation $\sum_{n=1}^{\infty} \mathbf{1}_{\{x > \epsilon n\}}$ as a function of x is plotted below.



We can write the infinite sum in the question as

$$\sum_{n=1}^{\infty} \mathbf{1}_{\{x/\epsilon > n\}}.$$

It counts the number of positive integers that are less than x/ϵ , which can be written as $\lfloor x/\epsilon \rfloor$. If we remove the floor operation, we obtain the upper bound x/ϵ .

(b) Let X denote a generic random variable whose distribution is the same as X_i . If we substitute x in the inequality in part (a) by X , and take expectation, we obtain

$$E\left[\sum_{n=1}^{\infty} \mathbf{1}_{\{X/n > \epsilon\}}\right] \leq \frac{E[X]}{\epsilon}.$$

We can exchange the order of expectation and integration on the left-hand side, because the indicator function are nonnegative and we can apply MCT. The expectation on the left is thus equal to

$$\sum_{n=1}^{\infty} E[\mathbf{1}_{\{X/n > \epsilon\}}] = \sum_{n=1}^{\infty} P(X/n > \epsilon).$$

Since the random variables X_i 's are identically distributed, with the same distribution as X , we can replace X by X_n to get

$$\sum_{n=1}^{\infty} P(X_n/n > \epsilon) \leq \frac{E[X]}{\epsilon}.$$

We now apply the first Borel Cantelli lemma to see that for each $\epsilon > 0$, we have

$$P\left(\frac{X_n}{n} > \epsilon \text{ i.o.}\right) = 0.$$

Hence, X_n/n converges to 0 almost surely.

(We note that the assumption of independence is not used. The conclusion holds for identically distributed but dependent random variables.)

Question 6. (15 points) For $n = 1, 2, 3, \dots$, let X_n be independent random variables, such that $P(X_n = 1) = 1/2 = P(X_n = -1)$. Let S_n denote the sum $X_1 + X_2 + \dots + X_n$, and define $X_0 = S_0 = 0$. The sequence of random variables $(S_n)_{n \geq 0}$ is a symmetric random walk.

(a) For integer $n \geq 1$, define Y_n to be the random variable $S_n^2 - n$, and denote the σ -algebra generated by $X_1, X_2, \dots, X_n$ by $\mathcal{F}_n$. We also define $Y_0 = 0$ and $\mathcal{F}_0 = \{\emptyset, \Omega\}$ for $n = 0$. Show that $(Y_n)_{n=0}^{\infty}$ is a martingale relative to the filtration $(\mathcal{F}_n)_{n \geq 0}$

(b) Given any two positive integer a and b , let T be the first time instance n such that $S_n = -a$ or $S_n = b$. This is a stopping time relative to the filtration $(\mathcal{F}_n)_{n=0}^{\infty}$. Prove that $E[T] = ab$.

(In lecture, we have shown that T is a stopping time, and $S_T = b$ with probability $a/(a+b)$ and $S_T = a$ with probability $b/(a+b)$. You can use these facts in your answer without repeating the proof.)

Solution. (a) We check the three defining conditions of martingale

(i) For all n , the random variables $X_1, \dots, X_n$ are integrable, and hence the expectation of the squared sum $(X_1 + X_2 + \dots + X_n)^2$ is finite.

(ii) For each n , Y_n is measurable with respect to $\sigma(X_1, \dots, X_n)$, because Y_n is a function of $X_1, \dots, X_n$.

(iii) For any $n \geq 0$,

$$\begin{aligned} E[Y_{n+1} | \mathcal{F}_n] &= E[(X_1 + \dots + X_n + X_{n+1})^2 - (n+1) | \mathcal{F}_n] \\ &= E[(X_1 + \dots + X_n)^2 + 2(X_1 + \dots + X_n)X_{n+1} + X_{n+1}^2 - (n+1) | \mathcal{F}_n] \\ &= (X_1 + \dots + X_n)^2 + 2(X_1 + \dots + X_n)E[X_{n+1} | \mathcal{F}_n] + E[X_{n+1}^2 | \mathcal{F}_n] - n - 1 \end{aligned}$$

Since $\sigma(X_{n+1})$ and $\mathcal{F}_n$ are independent, the conditional expectations $E[X_{n+1} | \mathcal{F}_n]$ and $E[X_{n+1}^2 | \mathcal{F}_n]$ are equal to $E[X_{n+1}]$ and $E[X_{n+1}^2]$, respectively. We thus obtain

$$E[Y_{n+1} | \mathcal{F}_n] = (X_1 + \dots + X_n)^2 + 2(X_1 + \dots + X_n) \cdot 0 + 1 - n - 1 = (X_1 + \dots + X_n)^2 - n = Y_n.$$

(b) We can apply the optional stopping theorem to the martingale $(Y_n)_{n=0}^\infty$, because the third condition in the theorem is satisfied. In fact, the expectation of T is finite, because, regardless of where the current location, if we move the right on $a+b$ consecutive steps, the random walk will stop. This happens with probability $1/2^{a+b}$. The expectation of T is no larger than $(a+b)$ times the expectation of a geometric distribution with success probability $1/2^{a+b}$. Hence, $E[T] \leq (a+b)2^{a+b}$. Moreover, the increment of Y_n is bounded by

$$|Y_{n+1} - Y_n| = |S_{n+1}^2 - n - 1 - S_n^2 + n| = |X_{n+1}^2 + 2X_{n+1}(X_1 + \cdots + X_n) - 1|.$$

Since $X_{n+1}^2 = 1$, the difference $|Y_{n+1} - Y_n|$ is bounded by $2|(X_1 + \cdots + X_n)| \leq 2\max(a, b)$.

When the random walk stops at time T , it is equal to $b^2 - T$ with probability $a/(a+b)$ and is equal to $a^2 - T$ with probability $b/(a+b)$. Hence,

$$\begin{aligned} 0 = E[Y_0] &= E[Y_T] = \frac{b}{a+b} E[(a^2 - T)] + \frac{a}{a+b} E[(b^2 - T)] \\ &= \frac{b}{a+b} a^2 + \frac{a}{a+b} b^2 - E[T] \\ &= ab \left(\frac{a}{a+b} + \frac{b}{a+b} \right) - E[T] \\ &= ab - E[T]. \end{aligned}$$

Since it is equal to 0, we must have $E[T] = ab$.

Question 7. (15 points) In this question, $(X_n)_{n=1}^\infty$ is a sequence of random variables converging to random variable X in probability.

(a) Prove that there exists a subsequence of X_n that converges to X almost surely.

(Hint: Let ϵ_k be a sequence of positive real numbers that converges to 0. By the assumption of convergence in probability, for any positive integer k , there exists an integer $N(k)$ such that $P(|X_n - X| \geq \epsilon_k) \leq 2^{-k}$ for all $n \geq N(k)$. The subsequence $X_{N(k)}$ is a candidate answer.)

(b) In general, we cannot deduce that X_n converges to X in the mean. However, this can be done with an extra assumption. Assume that there exists an integrable random variable Y such that $|X_n| \leq Y$ for all n . Show that X_n converges to X in the mean as $n \rightarrow \infty$.

(Hint: Show that it would contradict the dominated convergence theorem if X_n does not converge to X in L^1 . Start with assuming that X_n does not approach X in the mean. This means that there exists an $\epsilon > 0$ and a sequence $i_1, i_2, i_3, \dots$ in $\mathbb{N}$ such that $E[|X_{i_k} - X|] > \epsilon$ for all $k \geq 1$.)

Solution. (a) We want to show that for any real number ϵ between 0 and 1, the event

$$\limsup_k \{\omega : |X_{N(k)}(\omega) - X(\omega)| \geq \epsilon\}$$

has probability 0.

Comparing the sequence $(\epsilon_k)_{k \geq 1}$ with ϵ , since $\epsilon_k \searrow 0$, there exists an integer K such that $\epsilon_k < \epsilon$ for all $k \geq K$. Hence, for $k \geq K$, the event $\{|X_{N(k)} - X| \geq \epsilon\}$ has probability less than 2^{-k} . The sum of 2^{-k} for $k \geq K$ is finite. The infinite sum

$$\sum_{k=1}^{\infty} P(|X_{N(k)} - X| > \epsilon)$$

can be divided into two parts,

$$\sum_{k=1}^{K-1} P(|X_{N(k)} - X| > \epsilon) + \sum_{k=K}^{\infty} P(|X_{N(k)} - X| > \epsilon).$$

The first term is finite because it contains finitely many terms. The second term is finite because it is upper bounded by

$$\sum_{k=K}^{\infty} 2^{-k} = 2 \cdot 2^{-K}.$$

By BC lemma 1, we can conclude that $X_{N(k)}$ is converging to X almost surely as $k \rightarrow \infty$.

(A common mistake in part (a) is to jump to the conclusion by merely check that

$$\sum_{k=1}^{\infty} P(|X_{N(k)} - X| > \epsilon_k) \leq \sum_{k=1}^{\infty} 2^{-k} < \infty.$$

This not the same as the condition

$$\sum_{k=1}^{\infty} P(|X_{N(k)} - X| > \epsilon) < \infty,$$

in which the constant ϵ does not depend on k . Moreover, we need to show that the above condition holds for all $\epsilon > 0$ in order to prove almost sure convergence.)

(b) Let $i_1, i_2, i_3, \dots$ be a subsequence in $\mathbb{N}$ as defined in the hint. The subsequence of random variables $X_{i_1}, X_{i_2}, X_{i_3}, \dots$ also converge to X in probability.

By part (a), we can select a further subsequence of $(i_k)_{k=1}^{\infty}$ that converges to X almost surely. We apply dominated convergence theorem to this subsequence. Use a side result of DCT that the subsequence converges in L^1 to X (Theorem 7.5 in textbook). There is a subsequence $(i_{j_\ell})_{\ell=1}^{\infty}$ of $(i_k)_{k=1}^{\infty}$ such that

$$E[|X_{i_{j_\ell}} - X|] < \epsilon$$

for all sufficiently large ℓ .

This contradicts that $E[|X_{i_k} - X|] > \epsilon$ for all k .